CS471 - Lab 3

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Task 1: Build your first view function and corresponding URL mapping

In `apps/bookmodule/view.py` I added:

views.py M × wsgi.py urls.py .../bookmodule U

DjangoProjects > libraryproject > apps > bookmodule > views.py > ...

```python
from django.http import HttpResponse
from django.shortcuts import render

def index(request):
    # Task 1
    return HttpResponse("Hello, world!")
```

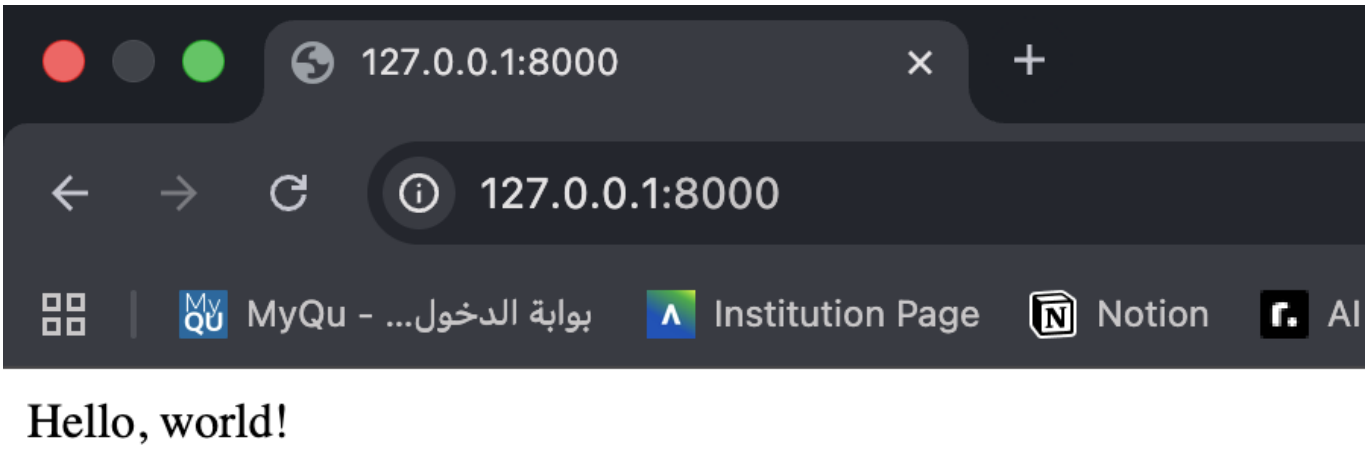
In `DjangoProjects/urls.py`, I added:

urls.py M ×

DjangoProjects > libraryproject > libraryproject > urls.py > ...

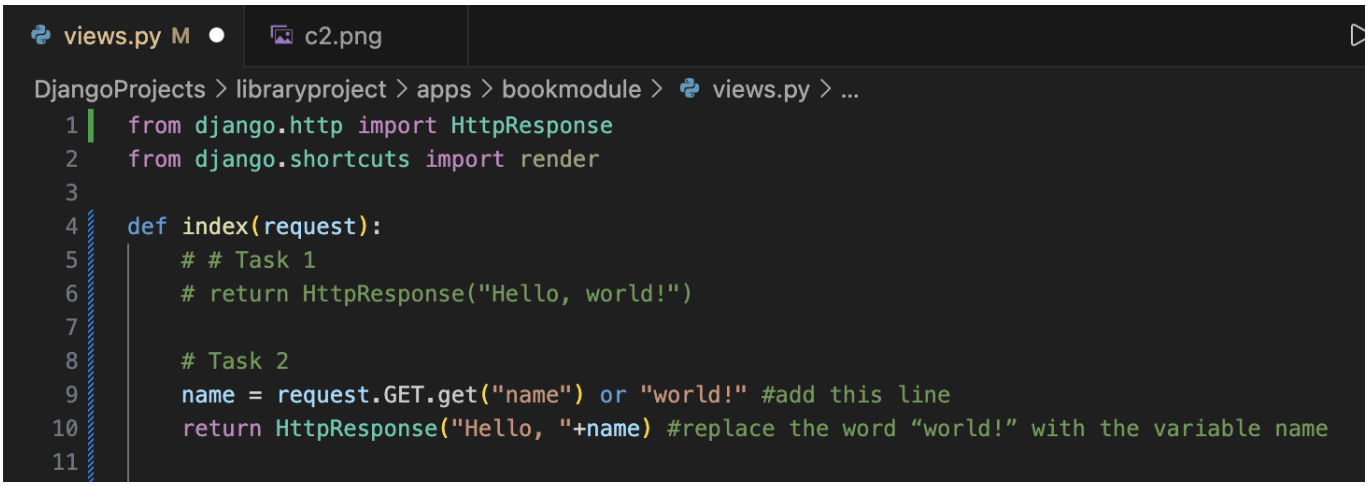
```python
import apps.bookmodule.views # Task 1

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', apps.bookmodule.views.index),
]
```

Then runned the server.

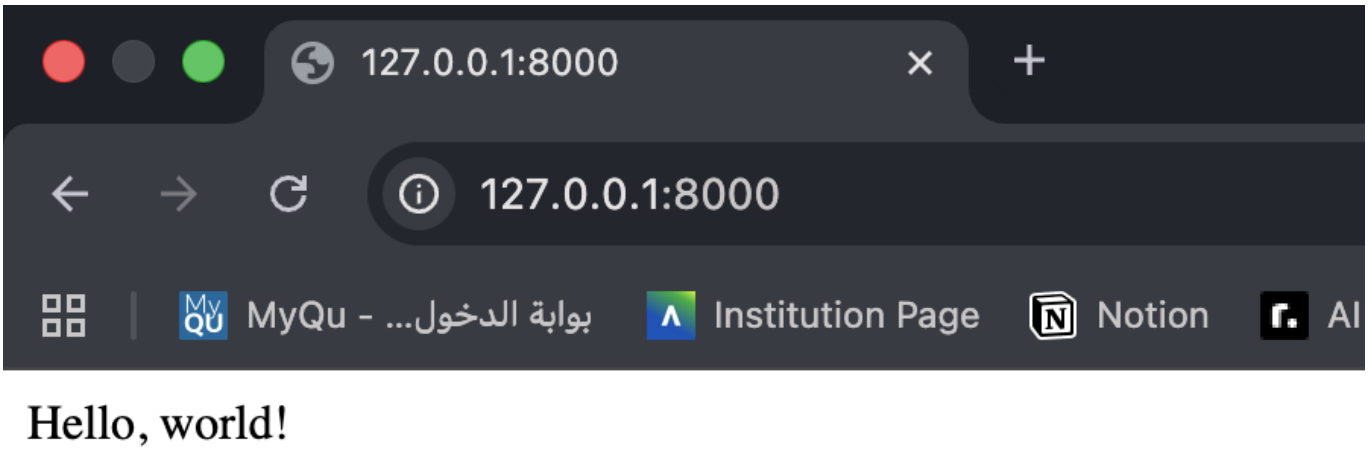


Task 2: Add parameters with HTTP requests

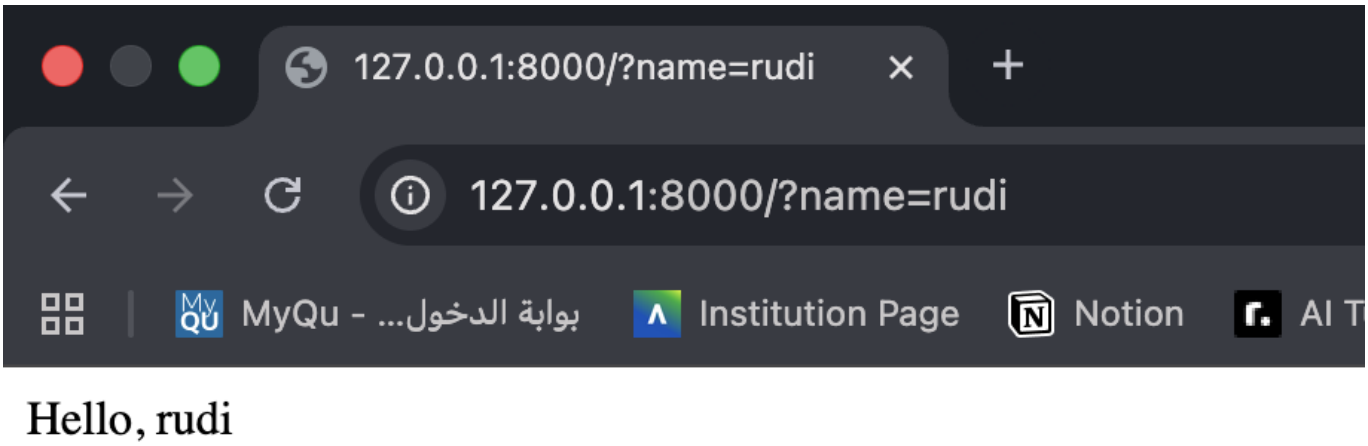
In `apps/bookmodule/views.py`:



`http://127.0.0.1:8000` Gives: “Hello, world!”



`http://127.0.0.1:8000?name=Rudi` Gives: “Hello, Rudi”



Task 3: Second view function and corresponding URL mapping with parameters within URL path

In `apps/bookmodule/views.py` added the second function:

```python
from django.http import HttpResponse
from django.shortcuts import render

def index(request):
    # # Task 1
    # return HttpResponse("Hello, world!")

    # Task 2
    name = request.GET.get("name") or "world!" #add this line
    return HttpResponse("Hello, "+name) #replace the word “world!” with the va

    # Task 3
def index2(request, val1 = 0):
    return HttpResponse("value1 = "+str(val1))
```

Then added a map to index2 view in the urlpatterns list in `DjangoProjects/urls.py`.

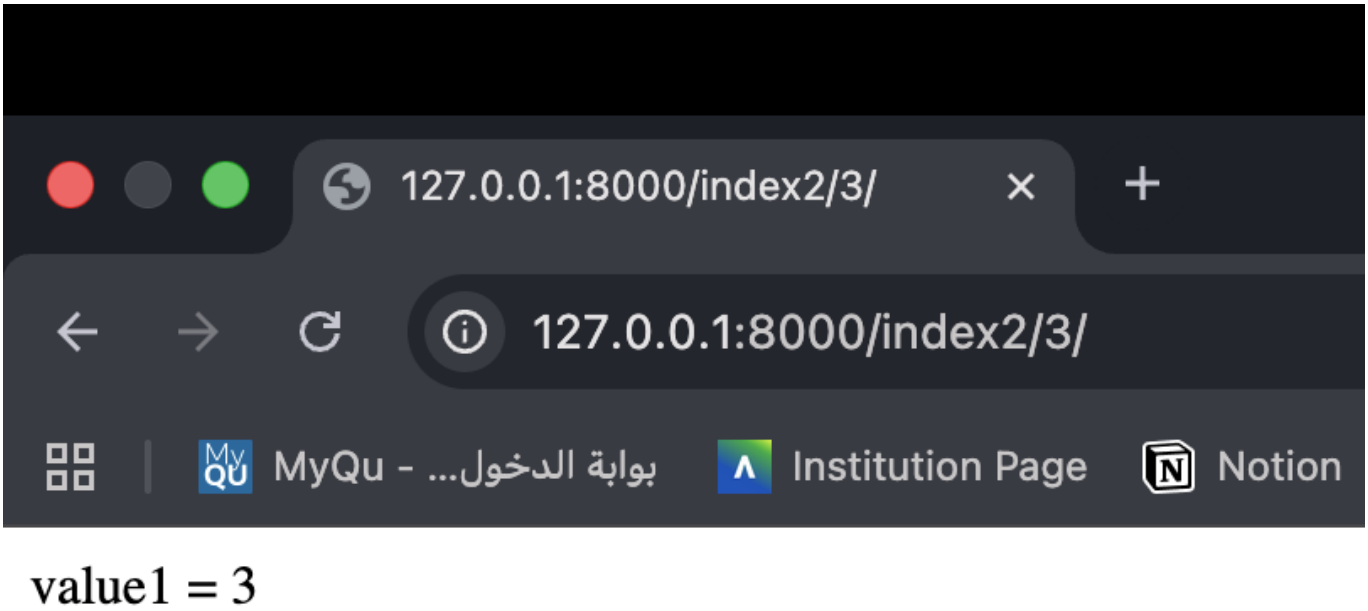
views.py M | settings.py M | Lab3.md | c4.png | urls.py M

DjangoProjects > libraryproject > libraryproject > urls.py > ...

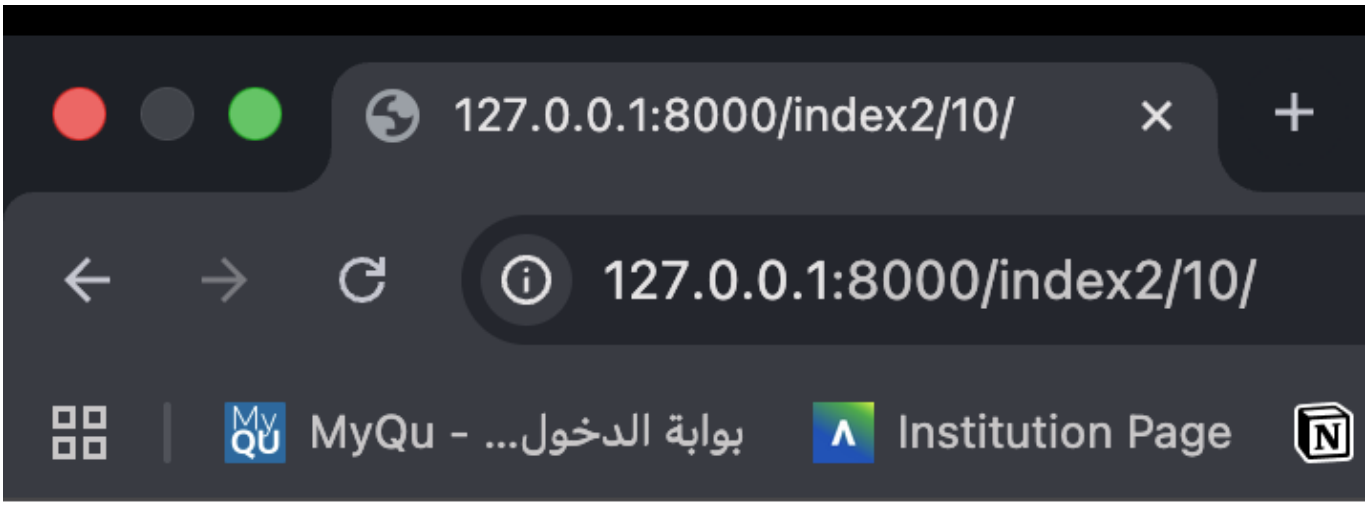
```python
from django.urls import include, path
import apps.bookmodule.views # Task 1

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', apps.bookmodule.views.index), # Task 1
    path('index2/<int:val1>/', apps.bookmodule.views.index2) # Task 3
]
```

Screenshots:

`http://127.0.0.1:8000/index2/3/` You should see “value1 = 3”



`http://127.0.0.1:8000/index2/10/` You should see “value1 = 10”

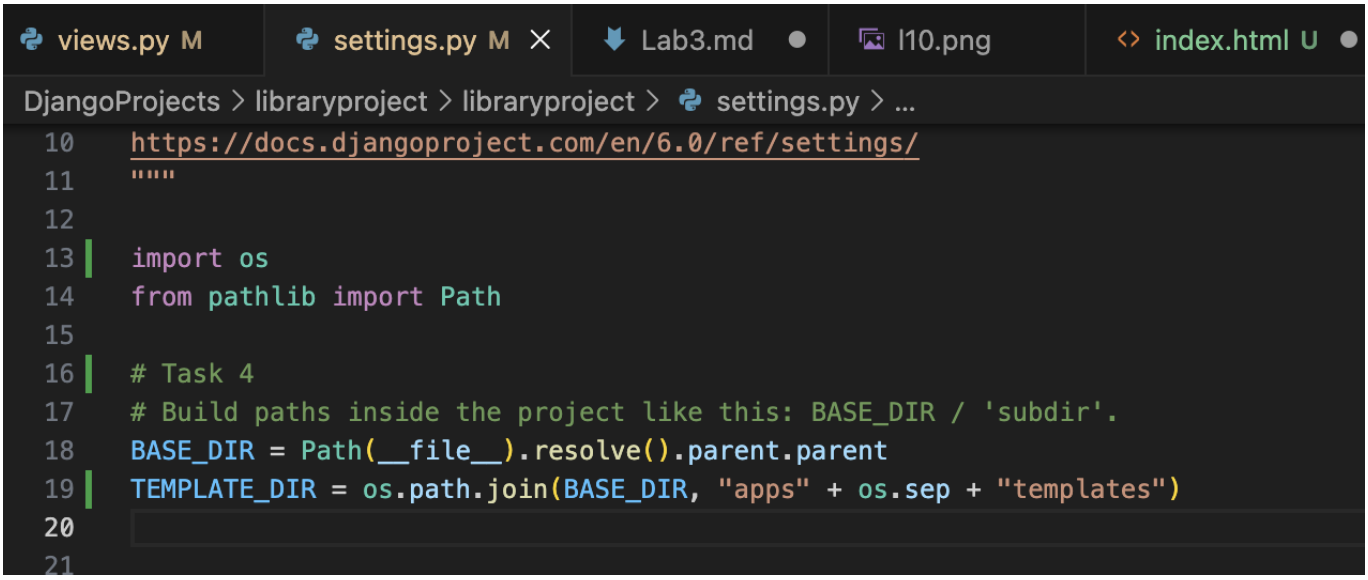


value1 = 10

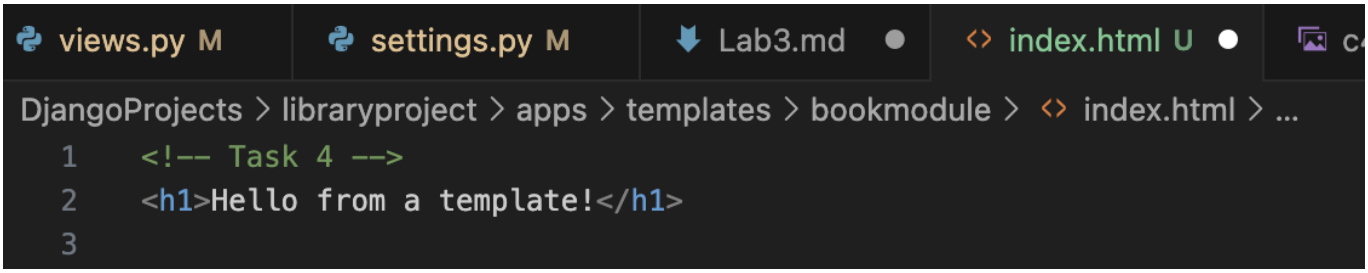
Task 4: Create a simple HTML template

Created a templates directory for bookmodule app.

Updated `settings.py` with template location.



Created a file with the name `index.html` within the templates directory.



Updated the index function, in `apps/bookmodule/views.py`, so that, instead of returning HttpResponse, it's returning a call to render:

```
views.py M ●   settings.py M   I9.png   Lab3.md ●   I10.png   index.ht
DjangoProjects > libraryproject > apps > bookmodule > views.py > ...
from urllib import request
from django.http import HttpResponse
from django.shortcuts import render

def index(request):
    # # Task 1
    # return HttpResponse("Hello, world!")

    # # Task 2
    # name = request.GET.get("name") or "world!" #add this line
    # return HttpResponse("Hello, "+name) #replace the word "world!" with the variable name

    # Task 4
    name = request.GET.get("name") or "world!"
    return render(request, "bookmodule/index.html")
```

Task 5: Rendering variables in the HTML template that processes a context

Updated `index.html` to render:

```
DjangoProjects > libraryproject > apps > templates > bookmodule > index.html >
<!-- Task 4 -->
<!-- <h1>Hello from a template!</h1>  -->

<!-- Task 5 -->
<h1>Hello from {{ name }}!</h1>
```

Updated the index function, in `apps/bookmodule/views.py`:

DjangoProjects > libraryproject > apps > bookmodule > views.py > index2

```python
from unicodedata import name
from urllib import request
from django.http import HttpResponse
from django.shortcuts import render

def index(request):
    # # Task 1
    # return HttpResponse("Hello, world!")

    # # Task 2
    # name = request.GET.get("name") or "world!" #add this line
    # return HttpResponse("Hello, "+name) #replace the word “world!” with the variable name

    # Task 4
    # name = request.GET.get("name") or "world!"
    # return render(request, "bookmodule/index.html")

    # Task 5
    name = request.GET.get("name") or "world!"
    return render(request, "bookmodule/index.html" , {"name": name}) #your render line

```

Task 6: Define patterns globally (DjangoProjects/urls.py) with specific urls.py file for each app/module

Created a `urls.py` for each app module (bookmodule and usermodule).

Moved urls of bookmodule to apps/bookmodule/urls.py, and include newly created urls.py in core/urls.py using include module.

`Djangoprojects/urls`:

DjangoProjects > libraryproject > libraryproject > urls.py > ...

```python
from django.contrib import admin
from django.urls import include, path
# import apps.bookmodule.views # Task 1

# urlpatterns = [
#     path('admin/', admin.site.urls),
#     path('', apps.bookmodule.views.index), # Task 1
#     path('index2/<int:val1>/', apps.bookmodule.views.index2) # Task 3
# ]


urlpatterns = [
    path('admin/', admin.site.urls),
    path('books/', include("apps.bookmodule.urls")), # include urls.py of bookmodule app
    path('users/', include("apps.usermodule.urls")) # include urls.py of usermodule app
]
```

`bookmodule/urls`:

DjangoProjects > libraryproject > apps > bookmodule > urls.py > ...

```python
from django.urls import path
from . import views

urlpatterns = [
    path('', views.index),
    path('index2/<int:val1>/', views.index2)
]
```

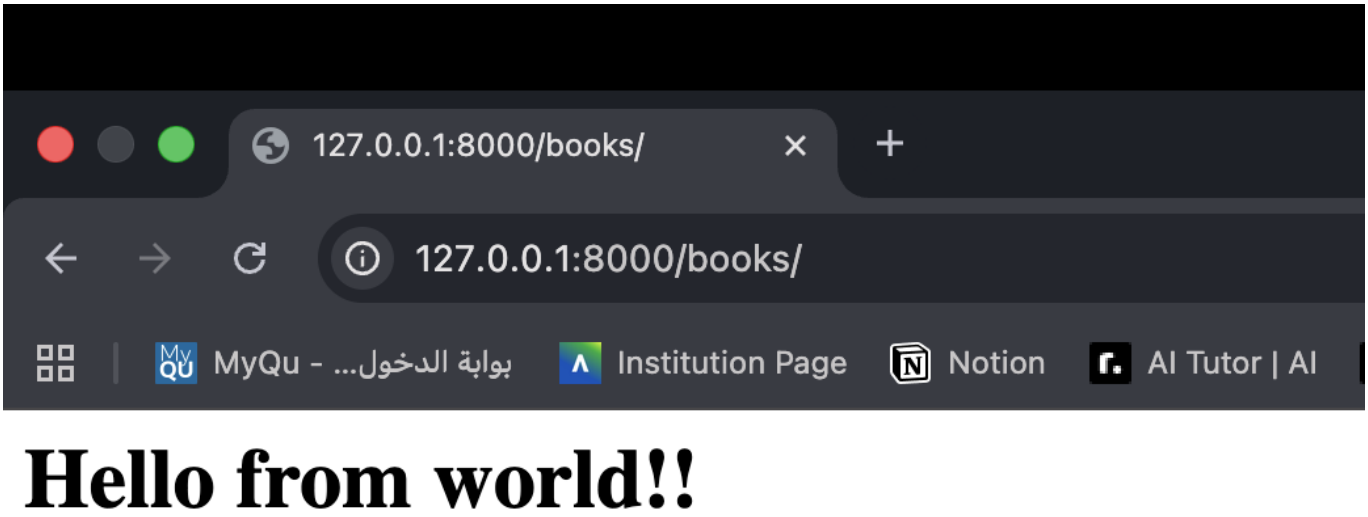
usermodule/urls:

DjangoProjects > libraryproject > apps > usermodule > urls.py > ...

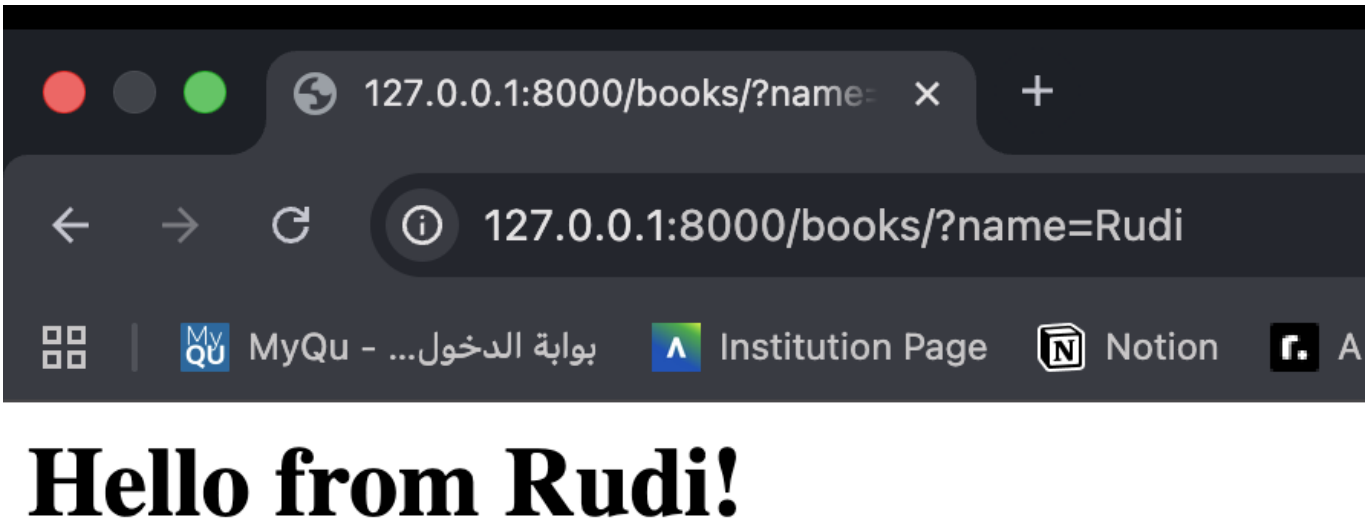
```python
from django.urls import path

urlpatterns = [
]
```

Screenshots:

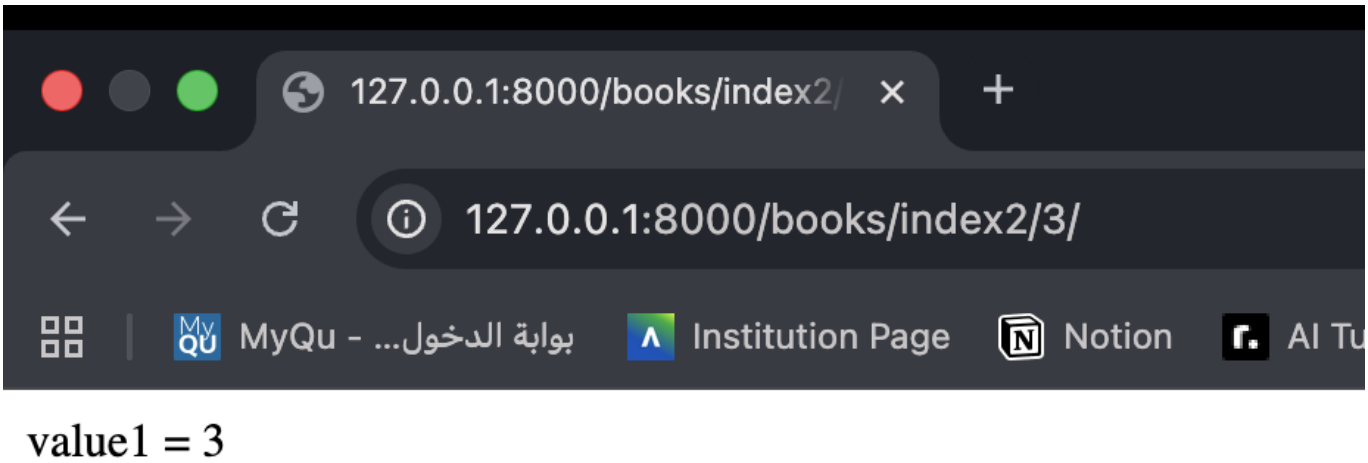
http://127.0.0.1:8000/books/ You should see “Hello, world!”



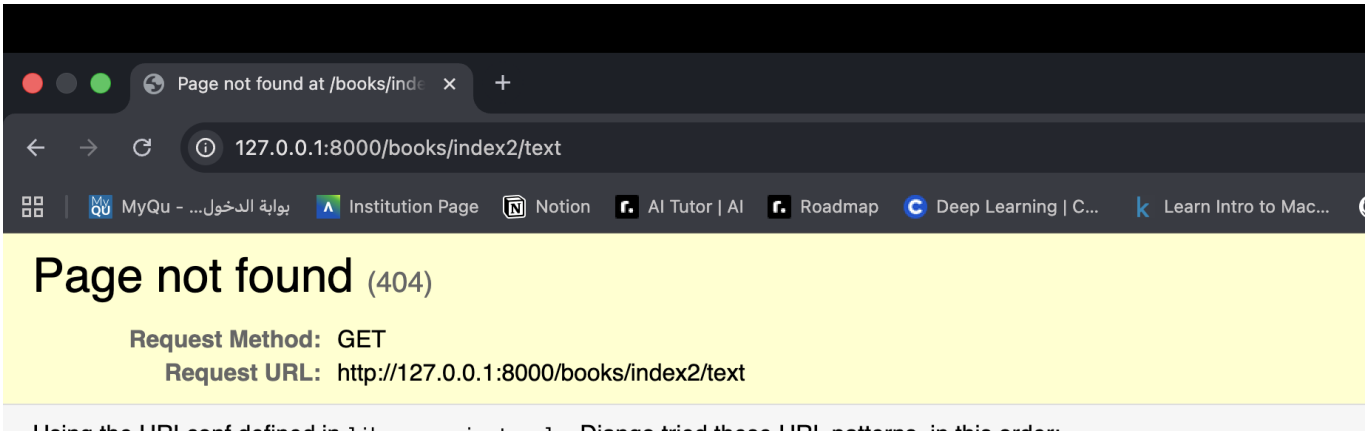
http://127.0.0.1:8000/books/?name=Rudi You should see “Hello, Rudi”



`http://127.0.0.1:8000/books/index2/3/` You should see “value1 = 3”



`http://127.0.0.1:8000/books/index2/text/` You should see “error, expected val1 to be integer”



Task 7: Create a URL, view, and HTML to display one book details

Created a new function at `bookmodule/views.py`:

```
views.py M ●   settings.py M   l9.png   Lab3.md ●   l3.png   l15.png
DjangoProjects > libraryproject > apps > bookmodule > views.py > ...
# Task 7
def viewbook(request, bookId):
    # assume that we have the following books somewhere (e.g. database)
    book1 = {'id':123, 'title':'Continuous Delivery', 'author':'J. Humble and D.Farley'}
    book2 = {'id':456, 'title':'Secrets of Reverse Engineering', 'author':'E. Eilam'}
    targetBook = None
    if book1['id'] == bookId: targetBook = book1
    if book2['id'] == bookId: targetBook = book2
    context = {'book':targetBook} # book is the variable name accessible by the template
    return render(request, 'bookmodule/show.html', context)
```

In `bookmodule/urls.py`, added the following path:

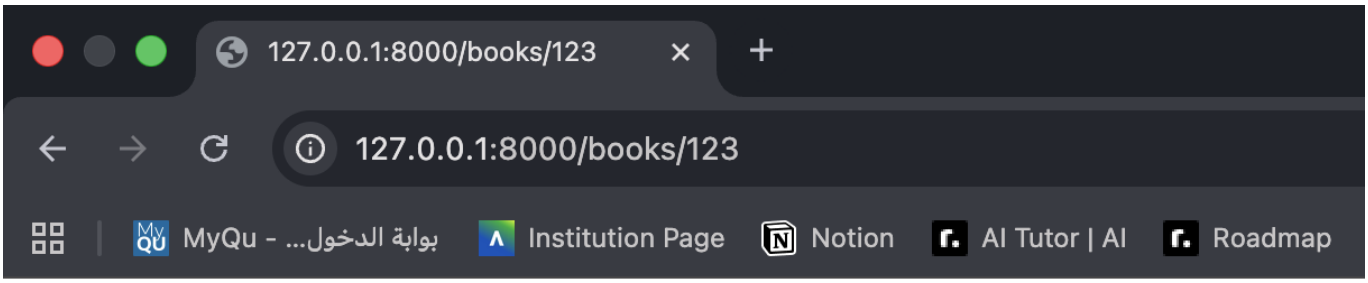
```
Lab3.md ●   l3.png   l15.png   l13.png   urls.py .../
DjangoProjects > libraryproject > apps > bookmodule > urls.py > ...
from django.urls import path
from . import views

urlpatterns = [
    path('', views.index),
    # path('index2/<int:val1>/', views.index2) # Task 6
    path('<int:bookId>', views.viewbook) # Task 7
]
```

Added `show.html`:

```
DjangoProjects > libraryproject > apps > templates > bookmodule > show.html >
<!-- Task 7 -->
<h4>ID: {{ book.id }}</h4>
<h4>Title: {{ book.title }}</h4>
<h4>Authors</h4>
<p>{{ book.author }}</p>
<h4>Description</h4>
<p>Description here</p>
```

Output:

`http://127.0.0.1:8000/books/123`:



ID: 123

Title: Continuous Delivery

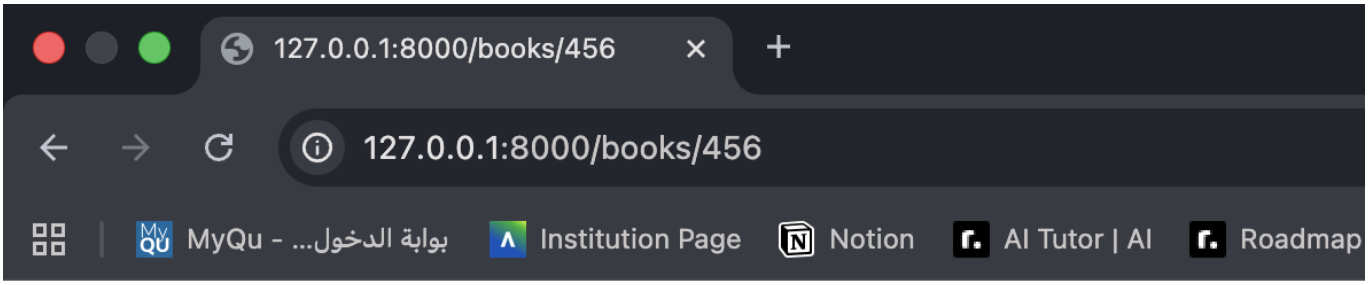
Authors

J. Humble and D.Farley

Description

Description here

`http://127.0.0.1:8000/books/456`:



ID: 456

Title: Secrets of Reverse Engineering

Authors

E. Eilam

Description

Description here